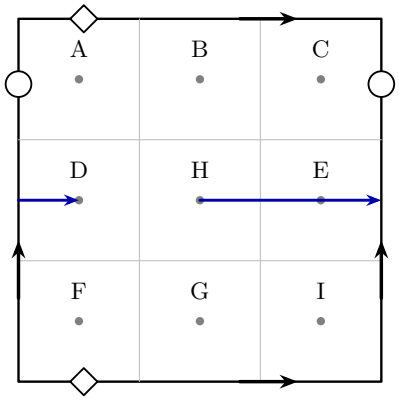
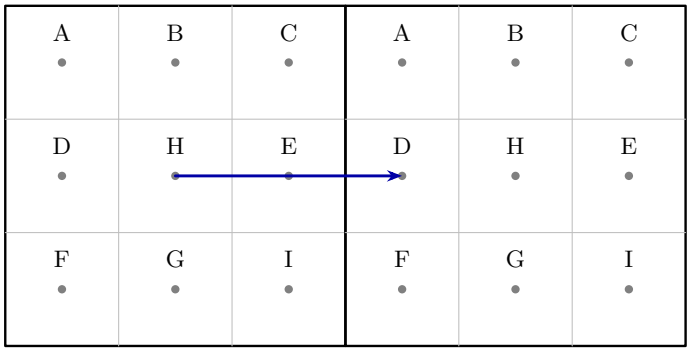
One arrow moves to the center of the next small square. Right meets left at the same height; top meets bottom at the same place across. Keep moving in the arrow direction. Repeated copies continue beyond the page.

input process output

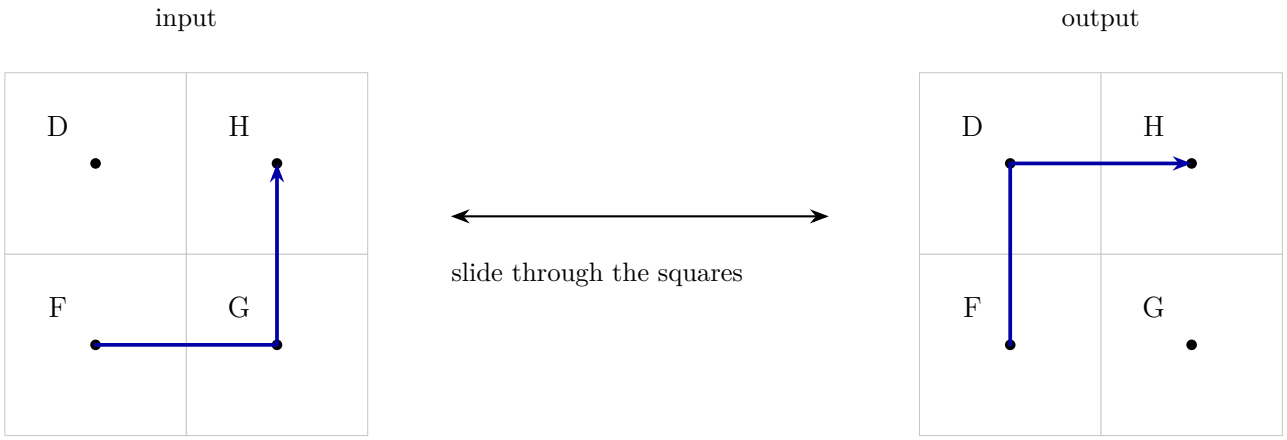


two steps right



original copy next copy

Journeys may move through the squares and cross themselves. Keep the start and finish of the whole journey at H, and keep its starting copy fixed. Add or erase two consecutive steps along the same segment in opposite directions. Within a journey, replace two adjacent sides of a square of cell centers by its other two sides, in either direction and any rotation.



Problem 4: Use tracing paper on the repeated map from page 3. Which trip can shrink all the way to staying at H? Show changes to the whole trip.

$\rightarrow \rightarrow \rightarrow \uparrow \uparrow \uparrow \leftarrow \leftarrow \leftarrow \downarrow \downarrow \downarrow$

$\rightarrow \rightarrow \rightarrow$

Problem 5: Change the first trip into the second using the allowed moves. Keep the same starting copy.

$\rightarrow \rightarrow \rightarrow \uparrow \uparrow \uparrow$

$\uparrow \uparrow \uparrow \rightarrow \rightarrow \rightarrow$

Problem 6: When can two trips from H back to H be changed into each other? Explain what their finishing copies decide, using the repeated map. Add copies if your journey leaves the page.

